Problem Statement and Goals
Software Engineering

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Table 1: Revision History

Date	Developer(s)	Change
September 24	Robert	Write problem statement and inputs/outputs
September 25	Nathan	Write goals and stretch goals
September 27	Aidan	Added stakeholders and environment
September 27	Keira	Write custom documents and reflection
September 28	Robert	Revise problem statement
September 28	All	Review document and fix formatting
October 10	Robert	Ensure goals support claims made in prob statement. Ensure all goals are measureable.

1 Problem Statement

1.1 Problem

Current AI inference markets are increasingly concentrated around a small number of large organizations [1]. As the artificial intelligence (AI) industry continues to grow, we become increasingly reliant on centralized providers for AI models, data, hardware, and compute. This centralization increases the risk of biased outputs because of the lack of diversity in training algorithms and data used by these companies [2]. Furthermore, the oligopilized nature of the AI market creates a risk of censorship and manipulation of AI outputs. The lack of transparency from these companies and the black-box nature of many AI models makes it difficult for consumers to verify the quality and reliability of their inferences. Lastly, the centralized nature of the compute infrastructure

used by these companies creates potential single points of failure. According to the 2026 Stanford AI Index, NVIDIA accounts for over 60% of total global AI compute capacity [3]. This concentration poses a significant risk of disruption to AI markets in the event of a single failure.

Our project is called DAIIM: Decentralized Artificial Intelligience Inference Marketplace. Our aim is to provide consumers with accurate AI predictions through a decentralized network of providers, reducing reliance on centralized solutions.

1.2 Inputs and Outputs

The inputs to the system consists of task definitions, inference requests, and network participation. Task definitions are specification of a problem to be solved via machine learning, including the expected input and output types. Inference requests ask the network to perform inference for a given task and include input data or some specification of where to find the input data. Network participation includes machine learning predictions for individual inference requests that are supplied to the network by an individual.

The system outputs aggregated inference results and maintains information about the network. The aggregated inference results include the final prediction for a given inference request, as well as information about the quality of the prediction.

1.3 Stakeholders

- **Task requesters**
  Task requesters are people who are requesting inferences to be made for a given task. These people would value the decentralized nature of our platform. For example, healthcare researchers would make requests to predict various regression inferences (such as biomarker levels) while still complying with privacy constraints. Similarly, decentralized finance platforms could make requests to predict various regression inferences (such as stock prices) while still complying with privacy constraints.

- **Inference providers**
  Inference providers are people with computing resources and models, and want to monetize their regression models. These people usually have specialized or powerful models that can be used to make predictions for a given task.

- **Data Providers**
  Data providers are people with data that can be used to train models for a given task. They would be interested in helping the inference providers improve their models by providing them with data to train on.

1.4 Environment

This software is compatible with any computer running a modern browser that is connected to wifi. There is no hardware component.

2 Goals

This project has a total of five primary goals.

2.1 Transparency and Decentralization

Enable inference consumers and model providers to participate in a transparent inference marketplace that does not rely on a single centralized entity. The marketplace's core logic, including payments, ownership, and settlement, will be deployed as smart contracts on a public blockchain. This ensures there are publicly verifiable records of requests, provider submissions, aggregated results, provider reputation etc. The system should require participation by multiple independent providers per inference, to reduce reliance on a narrow set of machine learning models and make the sources of inference more diverse. This goal will be measured by whether end-to-end testing demonstrate that a request can proceed through provider submission and settlement without a centralized coordinator, that the relevant marketplace state can be verified from on-chain records, and that at least two independent providers can participate in fulfilling requests.

2.2 Web3 Usability and Transparency

Provide a user-friendly Web3 experience that abstracts unnecessary blockchain complexity while maintaining transparency into the actions and transactions performed by the system. The application should clearly communicate the purpose and consequences of actions requiring user authorization, including the transactions and data being signed, while providing users with direct visibility into the corresponding on-chain transactions, their status, and relevant state changes. This goal will be evaluated by instructing users to complete representative Web3 tasks, such as authorizing a transaction and checking its status, then using a questionnaire to assess their understanding of the actions and ease of use.

2.3 Reliable Inferences

Enable users to receive accurate and reliable results by aggregating redundant predictions from multiple model providers rather than relying on a single prediction. Provider reputation scores will reflect observed performance and inform the aggregation, allowing contributions from providers with stronger records to have greater influence. As these scores are updated over repeated requests, the

network can adapt to provider performance and improve the quality of its aggregated results. Having multiple registered providers makes the network more reliable by allowing requests to be fulfilled by available providers when others are unavailable. Reliability will be measured by the request-completion rate: the percentage of submitted requests that receive an aggregated result within a predefined time limit. Tests will vary the number of participating providers while, and compare completion rates to assess how provider count affects reliability. The accuracy of aggregate network predictions will be evaluated against ground truths revealed at a later time. For example, for a weather forecasting task, the network's prediction will be compared with the actual weather conditions once the forecast period has elapsed.

2.4 Accessible Inference Experience

Design an accessible user interface that lets users without machine-learning experience submit the inputs required for a task and understand the resulting aggregated prediction and accompanying information. The interface should use clear signifiers for available actions, mappings between task inputs and controls, constraints that help prevent invalid submissions, and feedback that communicates request progress and results. This goal will be evaluated in usability tests with participants unfamiliar with machine learning. Participants will complete representative inference request and result interpretation tasks; evaluation measures will include task-completion rate, time on task, and user errors.

2.5 Model Provider Innovation

The network will encourage independent model providers to participate and innovate within the marketplace by providing opportunities to make their models available to consumers, compete for inference requests, and receive compensation for their services. The marketplace should create incentives for providers to improve the quality, performance, availability, and cost-effectiveness of their models, as models that provide greater value to consumers can attract more inference requests and generate greater potential revenue. If competition and these incentives encourage providers to improve their models, that improvement should be reflected in the network's aggregate predictions over time. This goal will be measured by tracking the network's aggregated predictions for a real inference task over repeated requests. As ground-truth outcomes become available, prediction quality will be assessed using a task-appropriate metric to determine whether the aggregated results improve over time.

3 Stretch Goals

This project has a total of two stretch goals, which outline additional features that could be implemented after the primary goals have been achieved.

3.1 Expanded AI Model and Inference Support

Extend the marketplace beyond regression to support additional categories of AI inference and more complex tasks. This is considered a stretch goal because supporting multiple model architectures and more complex inference workflows would require additional development beyond the core marketplace functionality established by the primary goals.

3.2 Private AI Inference

Enable inference on private or sensitive data while protecting that data from unauthorized access during execution through the use of Trusted Execution Environments (TEEs). This is considered a stretch goal because incorporating TEEs would introduce additional security, infrastructure, and integration requirements beyond those necessary to support the core marketplace and inference functionality.

4 Custom Documents (CDs)

4.1 Cognitive Walkthrough

A cognitive walkthrough will be performed using a wireframe prototype of the planned interface of the application from the point of view of the inference requesters. This will aid in mapping the entire process of DAIMM and generally smooth out any remaining ambiguity. The results of the cognitive walkthrough will provide essential insight into the required user feedback and how to make the UI accessible and learnable for the expected users to ensure the application is simple to understand and enjoyable to use.

4.2 Business Plan

A business plan will be developed, once a proof of concept has met expectations. The goal of the business plan is to detail how the decentralized AI marketplace can be implemented such that developers and stakeholders can be compensated, and will further explain the business strategy behind the tokenomics. The business plan would include a description of the problem and the market need, the market size and business model, the cost breakdown, the competition and positioning, and the execution plan and milestones.

Appendix — Reflection

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. **What went well while writing this deliverable?**

   Our team was incredibly agreeable and had no trouble communicating our ideas for the project or our concerns about the scope. The team worked smoothly to accomplish the requirements for the deliverable. Everyone cooperated with one another, and it was easy to split up the workload evenly without disagreements or conflict. We were able to easily schedule multiple group meetings, which facilitated group discussions that gave everyone a deeper understanding of the project.

2. **What pain points did you experience during this deliverable, and how did you resolve them?**

   There were varying levels of experience in the subject matter of ML, blockchains, inferences, and decentralized AI systems. Members who were less familiar with the content did some research on the subject matter, and members who were more experienced were extremely cooperative in answering questions in a non-judgmental and non-condescending manner. Notes of previous research performed by Robert were also shared as supplementary resources. Our team organized a couple of meetings to discuss the content in this document to ensure everyone was comfortable with the material and to encourage necessary discussion to determine the problem, users, goals, and scope of the project.

3. **How did you and your team adjust the scope of your goals to ensure they are suitable for a Capstone project (not overly ambitious but also of appropriate complexity for a senior design project)?**

   Our team discussed our skills and experience, and ideas for features of the application. We considered the timeline and budget, and ultimately we think we can reasonably accomplish the project over the next few terms given our skill and experience levels.

Initially, we brainstormed goals that we later deemed to be too ambitious for the scope of the project. For example, we had planned to try to implement multiple types of inferences. Upon further inspection, we realized that not only was this goal too solution-oriented, it was beyond what we could determine was reasonably feasible within the timeline of the project, considering the other features we are planning to implement. We instead included this as a stretch goal.

References

[1] Jai Vipra and Anton Korinek. Market Concentration Implications of Foundation Models. https://arxiv.org/abs/2311.01550, 2023.

[2] Esther Rolf, Theodora T. Worledge, Benjamin Recht, and Michael Jordan. Representation Matters: Assessing the Importance of Subgroup Allocations in Training Data. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139, pages 9040–9051, 2021. https://proceedings.mlr.press/v139/rolf21a.html.

[3] Stanford Institute for Human-Centered Artificial Intelligence. The 2026 AI Index Report. Technical report, Stanford University, 2026. https://hai.stanford.edu/ai-index/2026-ai-index-report.